

SAI HARSHA MUPPARAJU

☎ +1 (718)7723053 ✉ harshamupparaju9@gmail.com [in LinkedIn](#) [Website](#)

Education

New York University

2024 - May 2026 (Expected)

Master of Science in Computer Science. GPA:3.85/4

New York, USA

Relevant Courses: Algorithmic Machine Learning, Deep Learning, Data Structures and Algorithms, Cloud Computing and Big Data Systems, AR/VR, Artificial Intelligence

BITS Pilani

August 2019 – May 2023

Bachelor of Engineering in Electronics and Instrumentation, Minor in Data Science.

Goa, India

Publications

* denotes equal contribution

1. Perceptually Guided 3DGS Streaming and Rendering for Virtual Reality

Sai Harsha Mupparaju*, Yunxiang Zhang*, Kenneth Chen, Jenna Kang, Xinyu Zhang, Maito Omori, Kazuyuki Arimatsu, Qi Sun. *IEEE/CVF Winter Conference on Applications of Computer Vision (WACV 2026)*

2. Simple-RF: Regularizing Sparse Input Radiance Fields with Simpler Solutions [\[Paper\]](#) [\[Website\]](#)

Nagabhushan Somraj, Sai Harsha Mupparaju, Adithyan Karanayil, Rajiv Soundararajan

Conditionally Accepted to *ACM Transactions on Graphics (ACM TOG)*

3. Factorized Motion Fields for Fast Sparse Input Dynamic View Synthesis [\[Paper\]](#) [\[Website\]](#)

Nagabhushan Somraj, Kapil Choudhary, Sai Harsha Mupparaju, Rajiv Soundararajan

ACM SIGGRAPH (SIGGRAPH 2024)

Experience

Dwellci AI

June 2025 – August 2025

Machine Learning Dev Intern

New York, New York

- Developed a diffusion-based generative AI architecture for the Minimum Viable Product, enabling automatic floorplan generation from textual and visual room connectivity inputs.
- Designed and implemented a novel input-extraction pipeline that converts architectural bubble diagrams into model-ready spatial requirements, bridging traditional architectural workflows with generative design.

Indian Institute of Science

June 2023 – June 2024

Research Assistant

Bangalore, Karnataka

- Developed a framework that uses simpler augmented radiance models to regularize the depth of the main radiance model. [\[Website\]](#) [\[GitHub\]](#)
- Developed an explicit motion model as a factorized 4D representation that can take advantage of the spatio-temporal correlation of the motion field. [\[Website\]](#) [\[GitHub\]](#)

Samsung R&D India

July 2022 – December 2022

Software Development Intern

Bangalore, Karnataka

- Improved a voice assistant's ability to understand and answer indirect questions, and optimized its on-device performance for faster, more efficient inference.
- Built an intelligent recipe recommendation system powered by a structured food knowledge base, enabling more accurate and personalized suggestions.

Projects

Scalable Multi-Tenant Database | AWS, Dynamo, S3, Redis, CloudWatch, API Gateway, Lambda

- Built a scalable AWS multi-tenant DB system using SQLite shards with API-based schema management, queued migrations, and automated replication/failover across Lambda, DynamoDB, and S3.
- Implemented hot-cold storage tiering (EFS/S3), Redis caching, and automated rehydration to ensure low-latency access, high availability, and cost-efficient operation at scale.

Retrieval-Augmented Generation (RAG) Model for ROS2 | Python, PyTorch, Qdrant, MongoDB, HuggingFace

- Developed a ROS-focused RAG chatbot using a fine-tuned quantized Mistral model and a Qdrant-based retrieval pipeline, improving query resolution and response speeds by over 25–40%.
- Built scalable deployment with Docker, Gradio, and MongoDB, and optimized relevance and workflow automation using ClearML, boosting answer quality by 14% (MRR). [\[GitHub\]](#)

Technical Skills

Languages: Advanced: Python | Intermediate: Java, C, C++, JavaScript,

Tools/Frameworks: Advanced: PyTorch, Linux, Git | Intermediate: OpenCV, Unity, Blender